Ehsan Al Twal

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Skills

Software/Firmware: C, C++, Python, ROS 2, EtherCAT, Linux, RTOS, Git, Docker, Bash, Pytest, OpenCV, CNN, SLAM, DDS, NVIDIA Jetson, STM32, FOC, TypeScript, Doctest, J-Link, Cursor, MATLAB, Simulink, Java, C#, Assembly, VHDL, React
Electrical: Electrical Systems Design, PCB Design (Altium), BLDC Motor Control, ARM Microcontrollers (STM32, AX58400, Teensy 4.0), FPGAs (De1-SoC), Debugging & Soldering, SPI, I2C, CAN, RF, EtherCAT

Education

University of British Columbia Sep 2021 – Expected Dec 2026
B.A.Sc. (Bachelor of Applied Science) in Engineering Physics *Vancouver, Canada*
- Specialized in Robot Software Engineering

Professional Experience

Robot Software Engineer Dec 2025 – Present
Sanctuary AI *Vancouver, Canada*
- Redesigned hand command smoothing as a reusable **ros2_control** chain stage with a realtime-safe consume-and-clear handshake, decoupling external command cadence from the 1 kHz control loop after ruling out five alternative protocols via hardware experiments.
- Implemented a custom fault publisher and recovery controller to ensure safe fault recovery for bimanual dual **UR** robotic systems.
- Architected "MockCAT," an **EtherCAT** slave simulator that enables the software team to develop software and emulate full robotic hand networks without hardware.
- Deployed an **IgH EtherCAT** master on an **NVIDIA Jetson Orin** to control hydraulic robot hands, achieving stable 1 kHz control loops.
- Benchmarked real-time control paths across computing architectures, isolating **DDS** middleware latency to optimize the hand controller's network topology.

Firmware Engineer May 2025 – Dec 2025
Sanctuary AI *Vancouver, Canada*
- Developed robot's hydraulic power unit firmware on **AX58400**: system monitoring, safety interlocks, fan control, ECAT communication, and dual-core IPC.
- Tripled team throughput by building a local **EtherCAT** dev/test suite GUI, replacing shared Speedgoat resource, thus parallelizing development. Added a **Pytest** tool automating 75 PDO tests/sec.
- Developed a full-stack tactile data pipeline (firmware, **EtherCAT** Master and Slave, and **ROS 2** bridge) to stream sensor data and enable live teleoperation haptics.

Electrical Engineer / Firmware Engineer May 2024 – Dec 2024
Sarcomere Dynamics *Vancouver, Canada*
- Designed an in-house **STM32**-based **BLDC** motor driver PCB in **Altium**, covering power regulation (24 V to 5 V/3.3 V), the motor driver IC, an onboard magnetic encoder, and motor Hall sensor inputs.
- Developed a **BLDC** motor driver library in **C** for **STM32** microcontrollers, handling PWM generation, sensor feedback, and current sensing.
- Implemented **field-oriented control (FOC)** for brushless DC motors with position, velocity, and torque control loops.

Projects

Autonomous Quadruped Guide Dog
- Engineered an autonomous **C++/ROS 2** navigation stack on **Nvidia Jetson**, fusing **FAST-LIO** for 10 Hz real-time 3D odometry and mapping with a custom path-planning module to continuously generate optimal, collision-free routes with under 150 ms latency.
- Designed a custom hierarchical **C++** motion controller that fused a high-level **Pure Pursuit** state machine with a positional and angular **PD** heading controller, completely eliminating the need for basic PID fallbacks, to execute trajectory tracking with a positional accuracy of 2 cm.
- Deployed a real-time **machine learning** inference pipeline that predicted the trajectories of dynamic obstacles 2 seconds into the future to proactively trigger evasive path-planning maneuvers.